d-Matrix®

One Abstraction, Many Transports: Push-Mode Communication from Chiplet to Rack

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2026 OCP Global Summit

October 14th, 2026

San Jose, California

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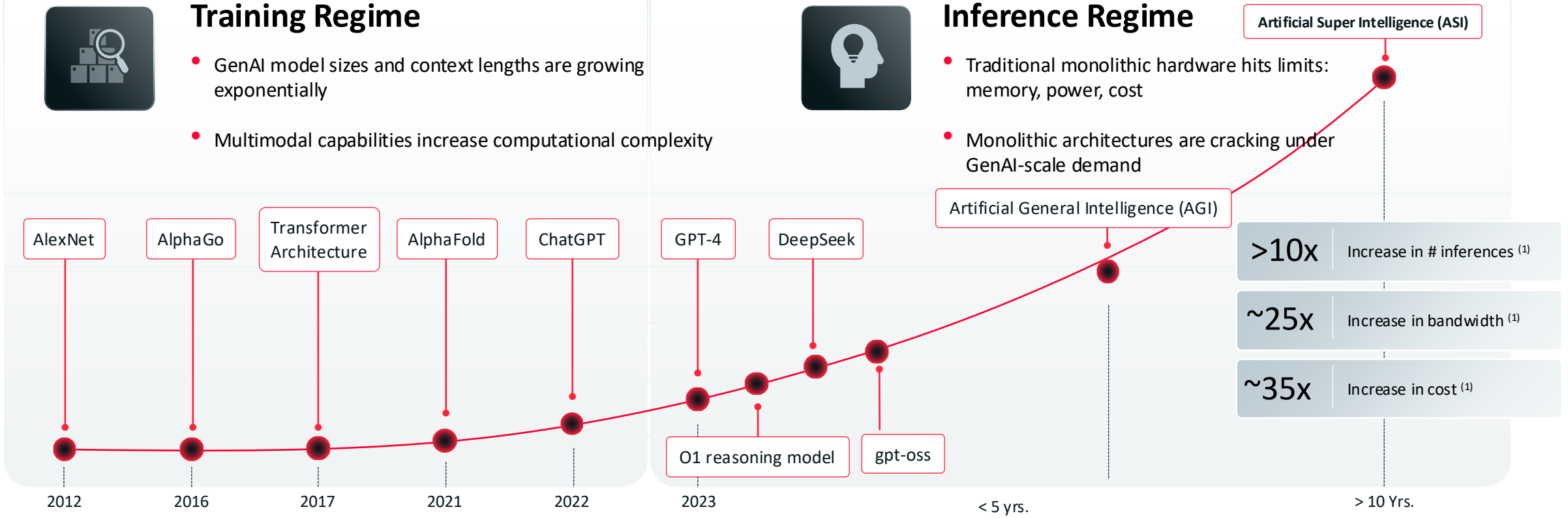
Rise of the Inference Regime

Training Regime

- GenAI model sizes and context lengths are growing exponentially
- Multimodal capabilities increase computational complexity

Inference Regime

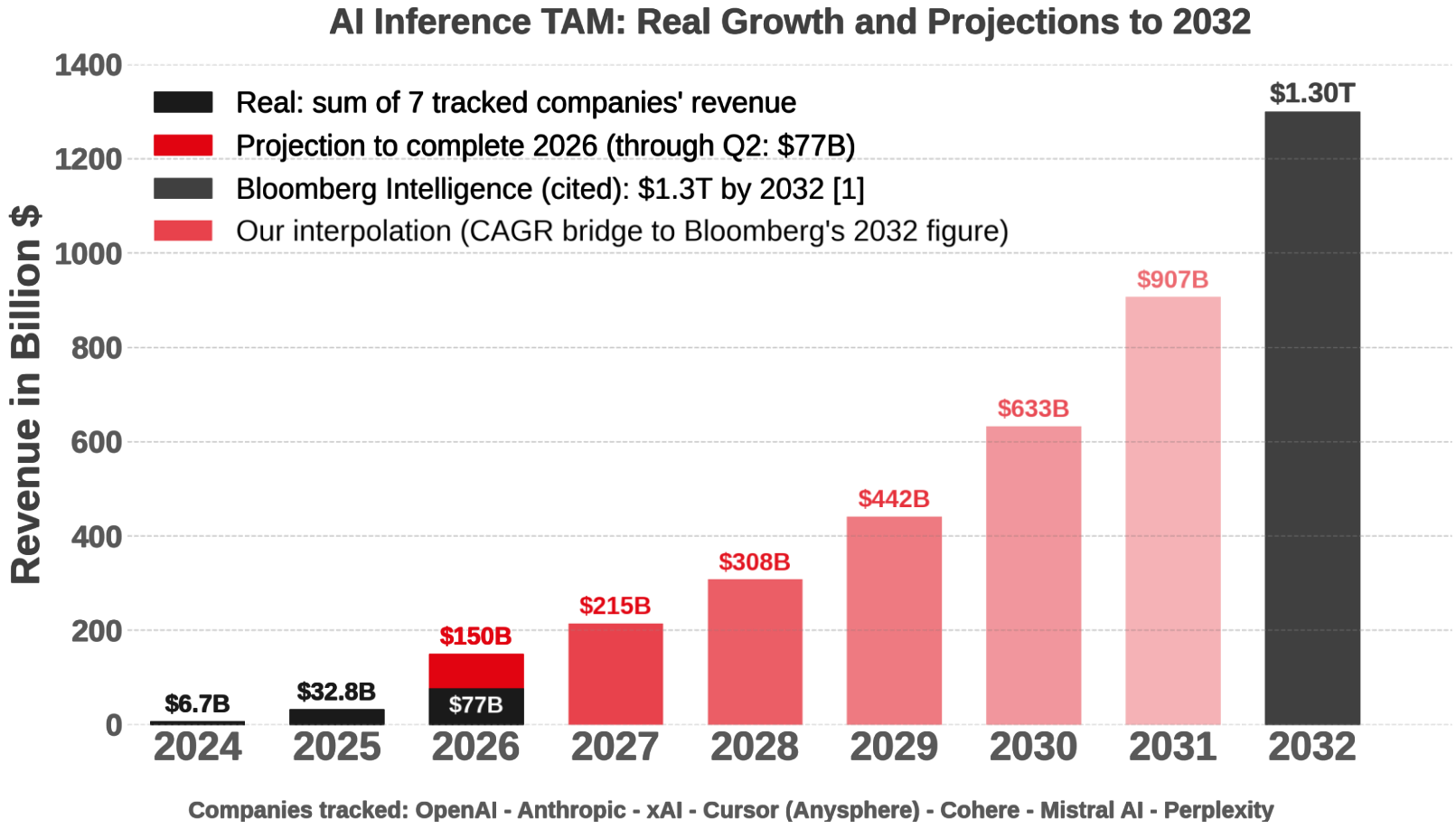
- Traditional monolithic hardware hits limits: memory, power, cost
- Monolithic architectures are cracking under GenAI-scale demand



One shot → Multi-turn → Reasoning → Agentic …

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LLM Inference Market



- LLM Inference revenue growth > 3x for 2 consecutive years.
- Bloomberg Intelligence : Generative AI inference market should grow beyond $1 Trillion by 2032 (32% CAGR).

Capturing this market isn't just about model quality — it's about serving economics.

[1] Bloomberg forecast of Inference market

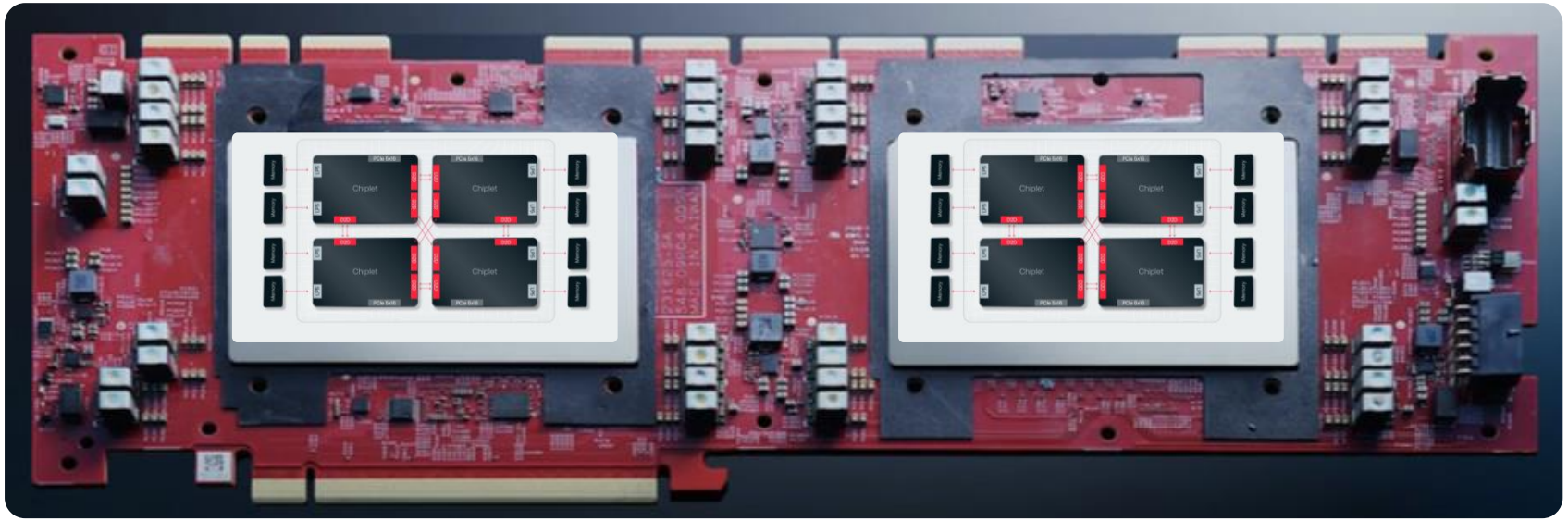
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Corsair

8 Chiplets
TSMC 6nm

2400 TFLOPS MXINT8
9600 TFLOPS MXINT4

PCIe: 128 GB/s
DMX Bridge: 512 GB/s



Deliver Inference with sub-millisecond latency

PCIe Gen5
600W TDP

2GB SRAM
Memory 150 TB/s

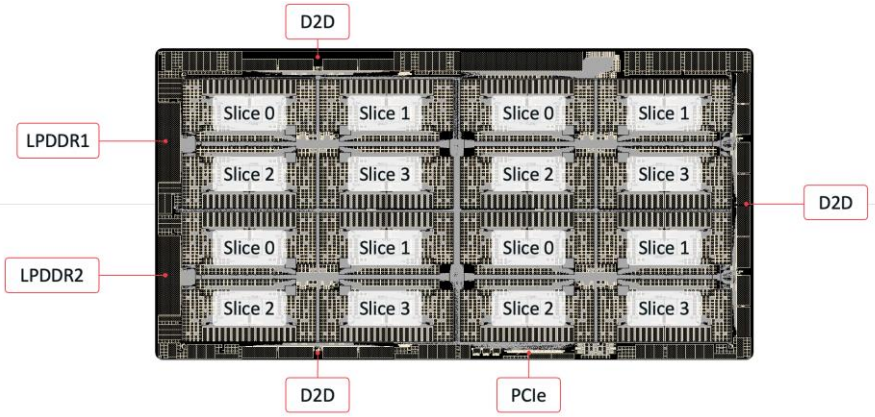
256GB LPDDR5
400 GB/s

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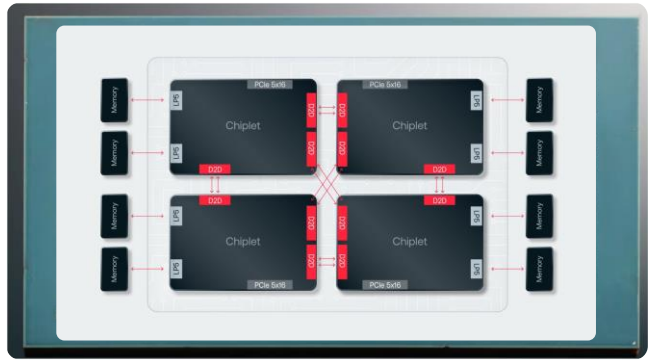
d-Matrix Connectivity Quadrant

(Different Transports and Push Mode Communication)

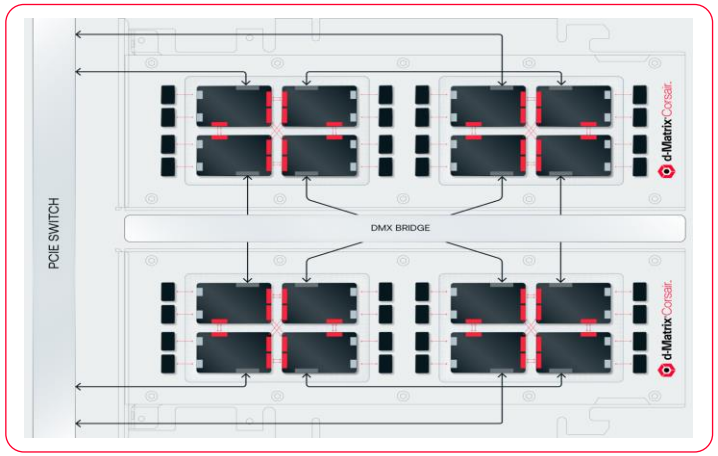
Chiplet (NOC)



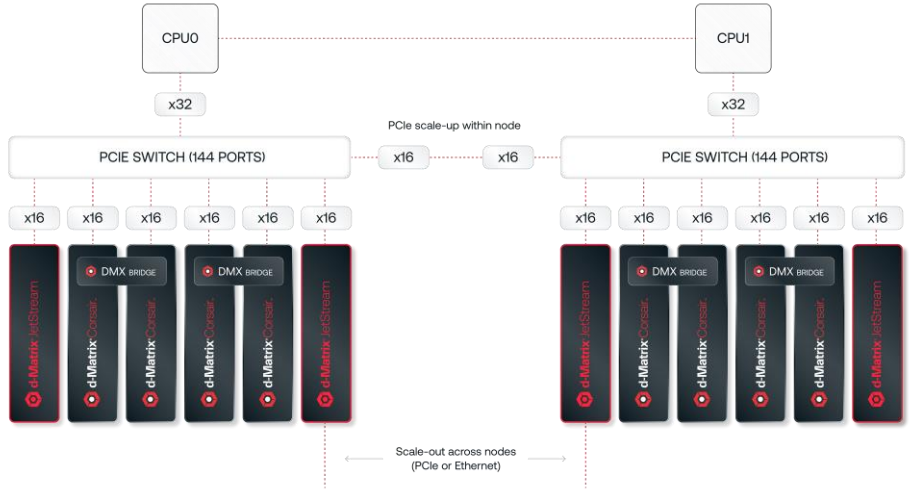
Package (D2D)



Card (TP-1 and TP-2) (PCIE)



Node (PCIE)



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d-Matrix JetStream™

Purpose-built I/O for AI Inference Scaling

PCIe Gen5
150W TDP



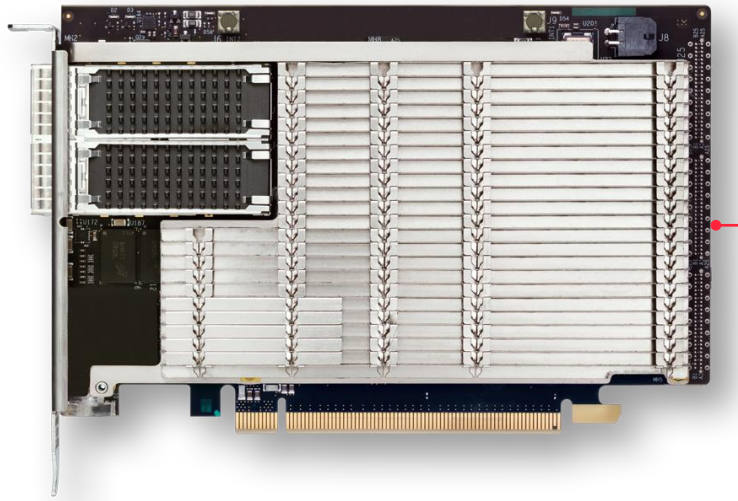
IEEE 802.3
Ethernet

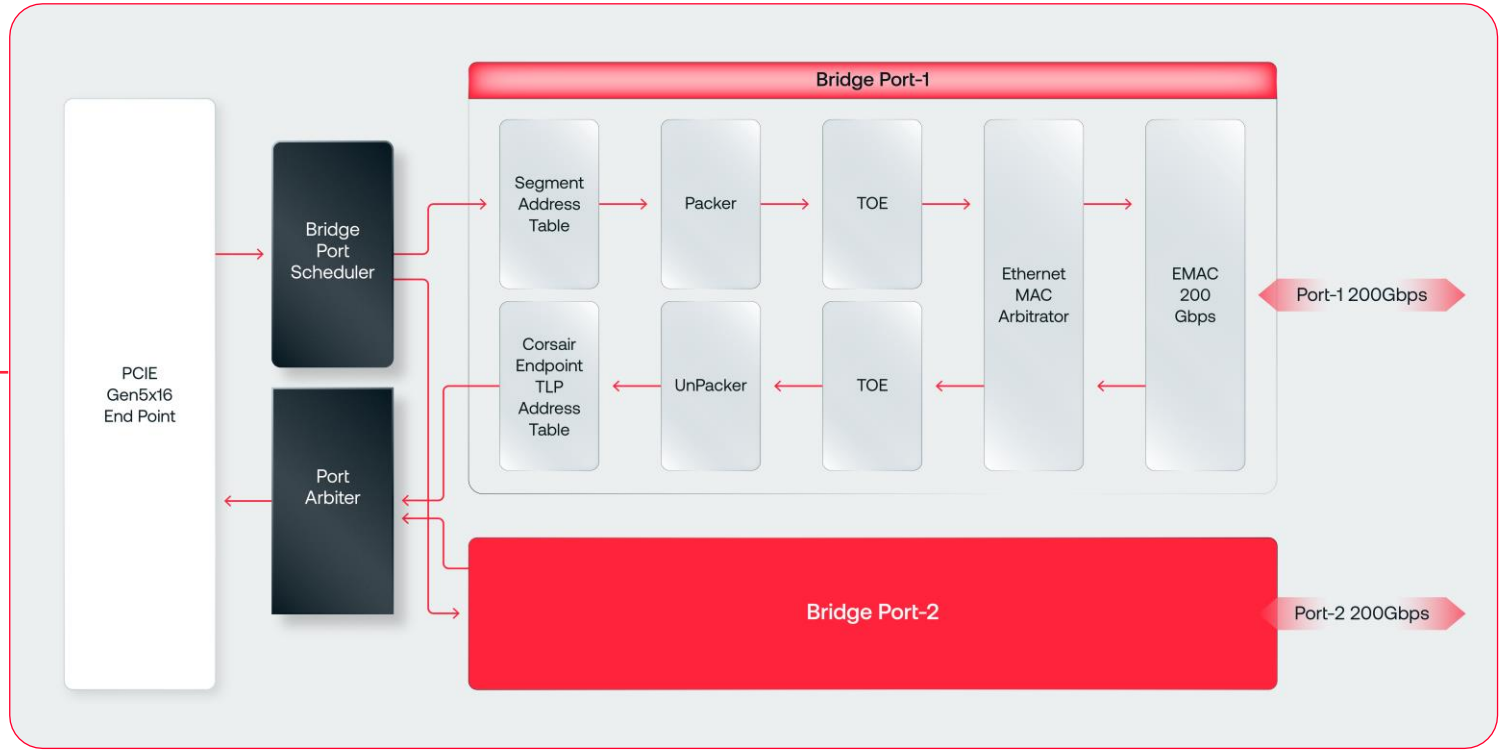
Latency 2us

400 Gbps

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JetStream NIC





- ✓ Transport PCIE semantics over Ethernet
- ✓ Scale communication semantics from intra-chiplet to inter-node
- ✓ Lower latency vs RoCEv2 (graph-to-graph)

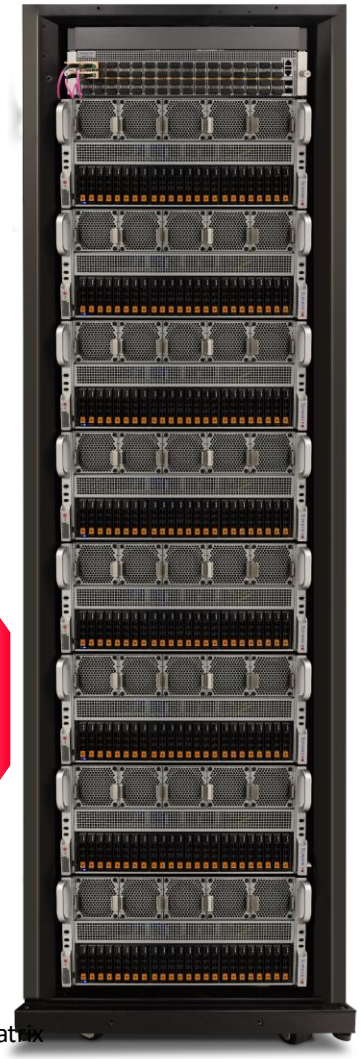
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SquadRack™

Open Rack-scale Reference Solution Purpose-Built To Speed Up Time to Inference









Reference Solution
Built in Collaboration With

SUPERMICRO ARISTA BROADCOM

Air-cooled
Open Standards Based
No vendor lock-in
Customizable to fit inference demand
In Existing Datacenters

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SquadRack™

d-Matrix Squadron™

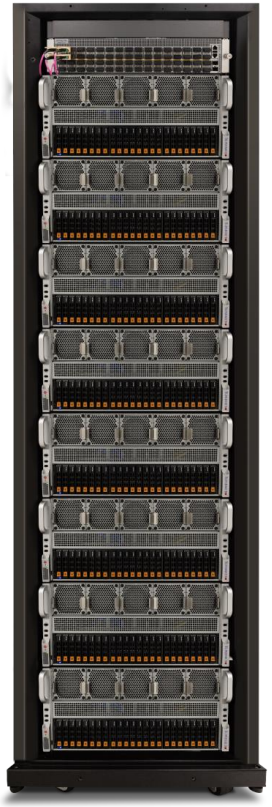
Clusters Scaling to 1,000+ XPUs for Ultra-Low-Latency AI

Clip, Whisper
GPT-OSS-20B



Half-rack

GPT-OSS-120B



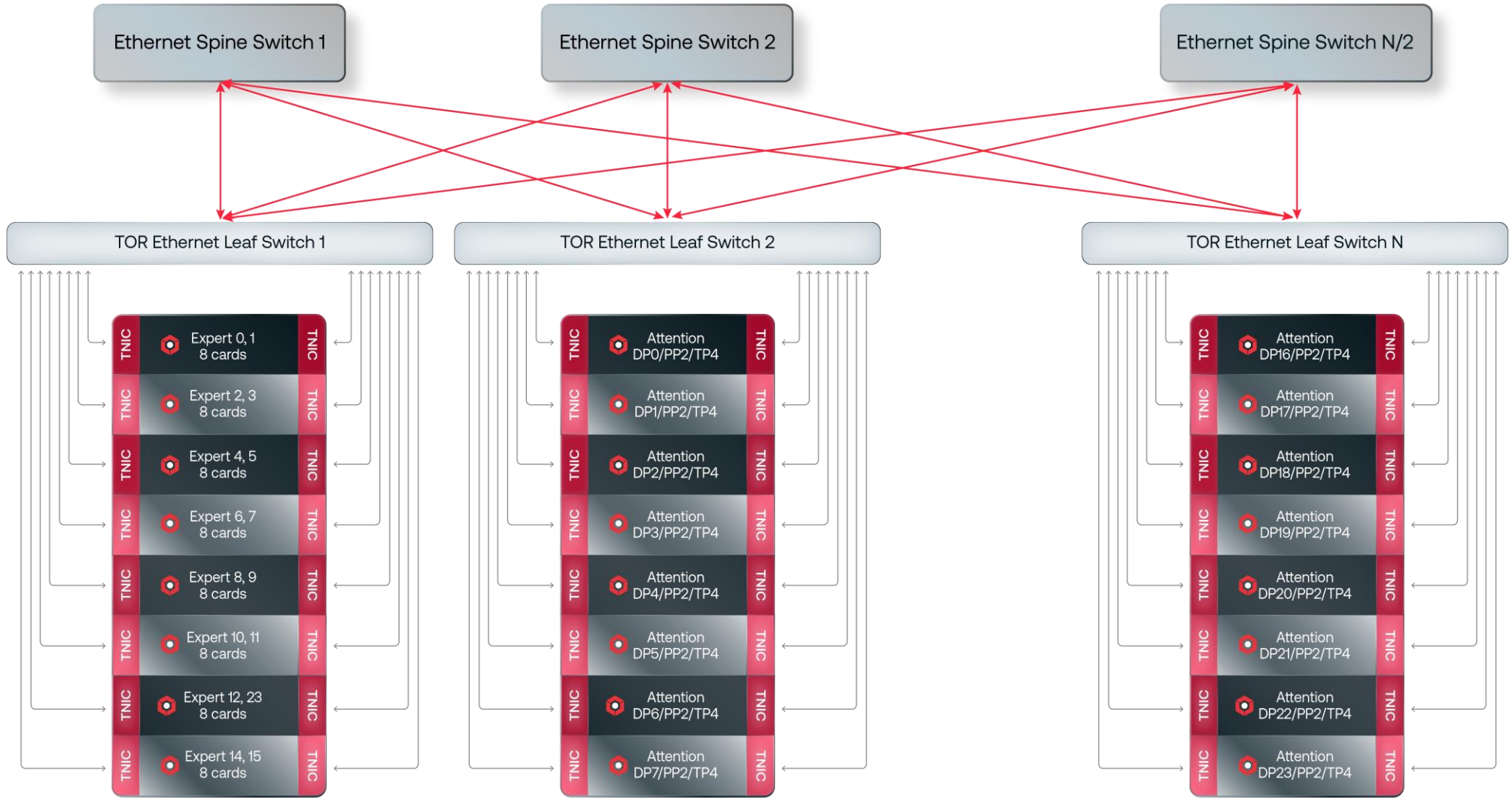
Full-rack



- Up to 16 SquadRacks
- Scales to 1,024 Corsair XPUs
- Megawatt-level compute
- 1st Squadron in Arizona for internal R&D
- Available now

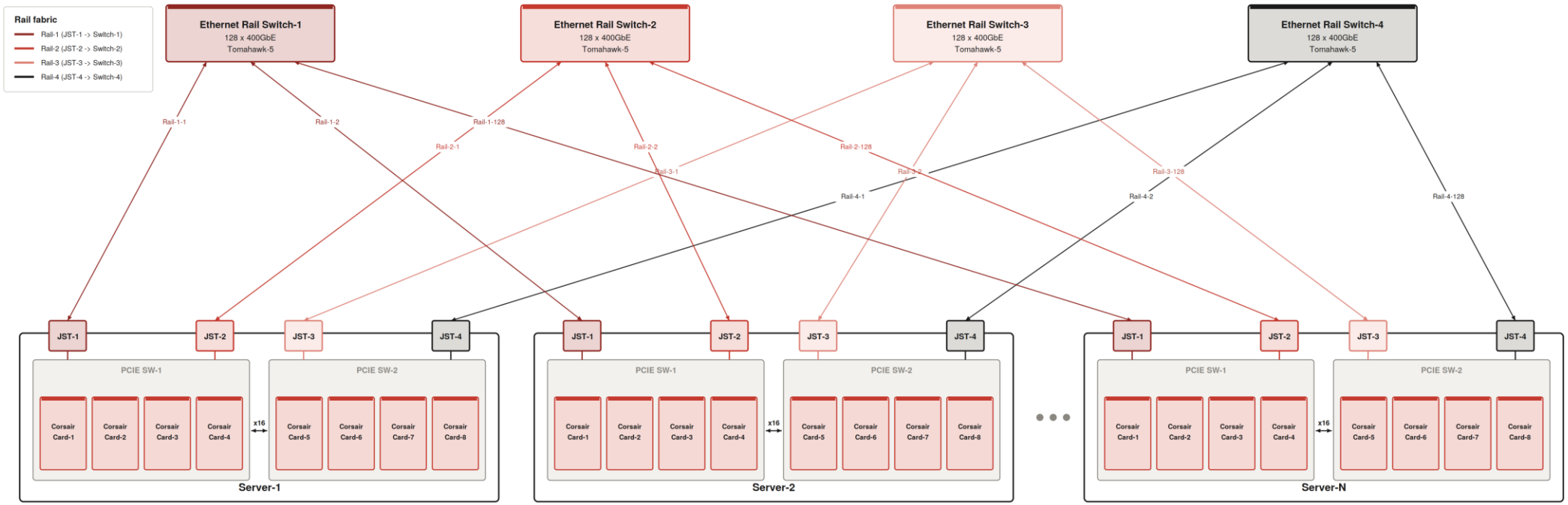
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Rack Level Scale-Out: Multi-node and Multi-Rack



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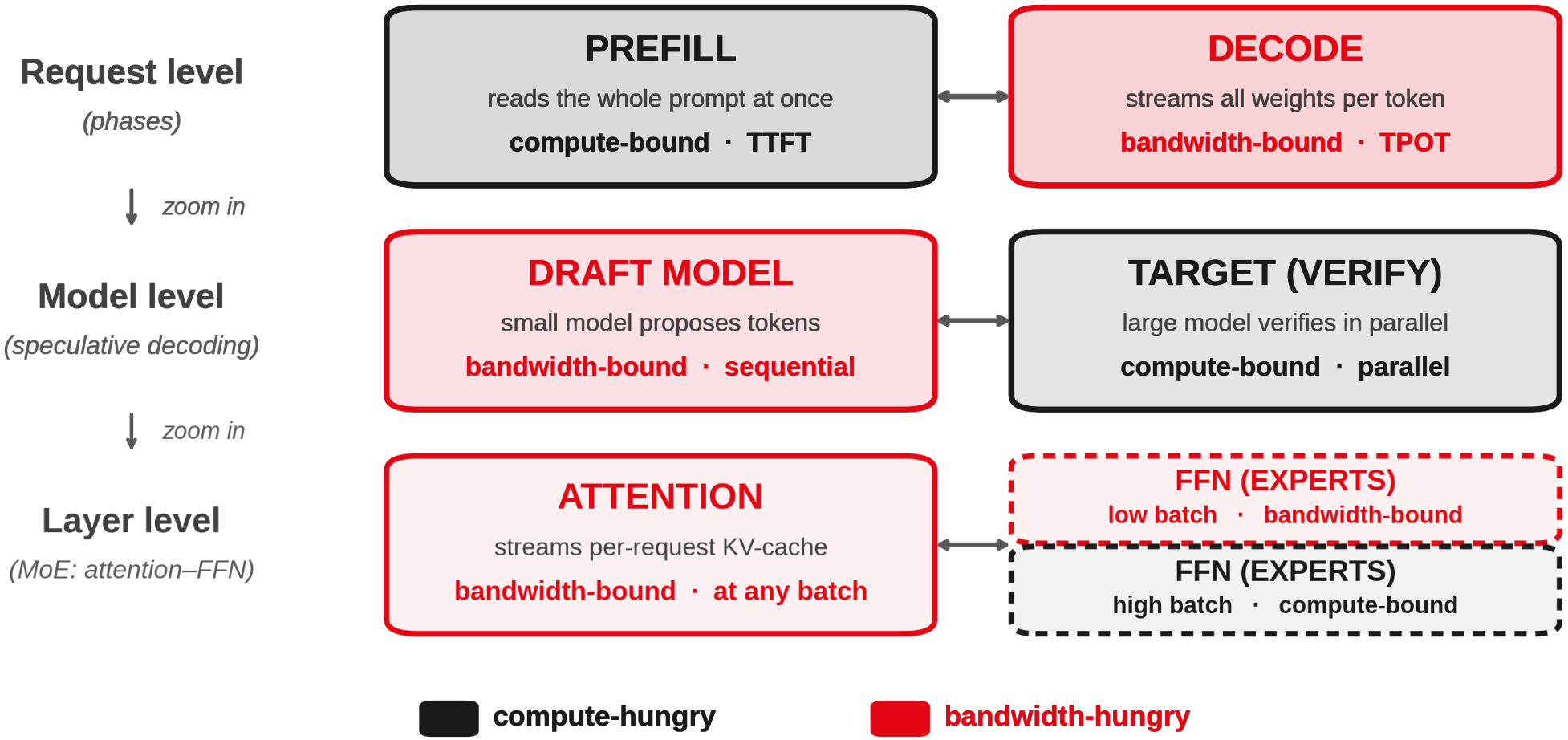
Rail Only Topology for Low Latency



Use TP/PP/EP inside a Node
Use PP/EP between Nodes through Ethernet Switches
Use Push Mode to communicate across rails

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Inference is not monolithic



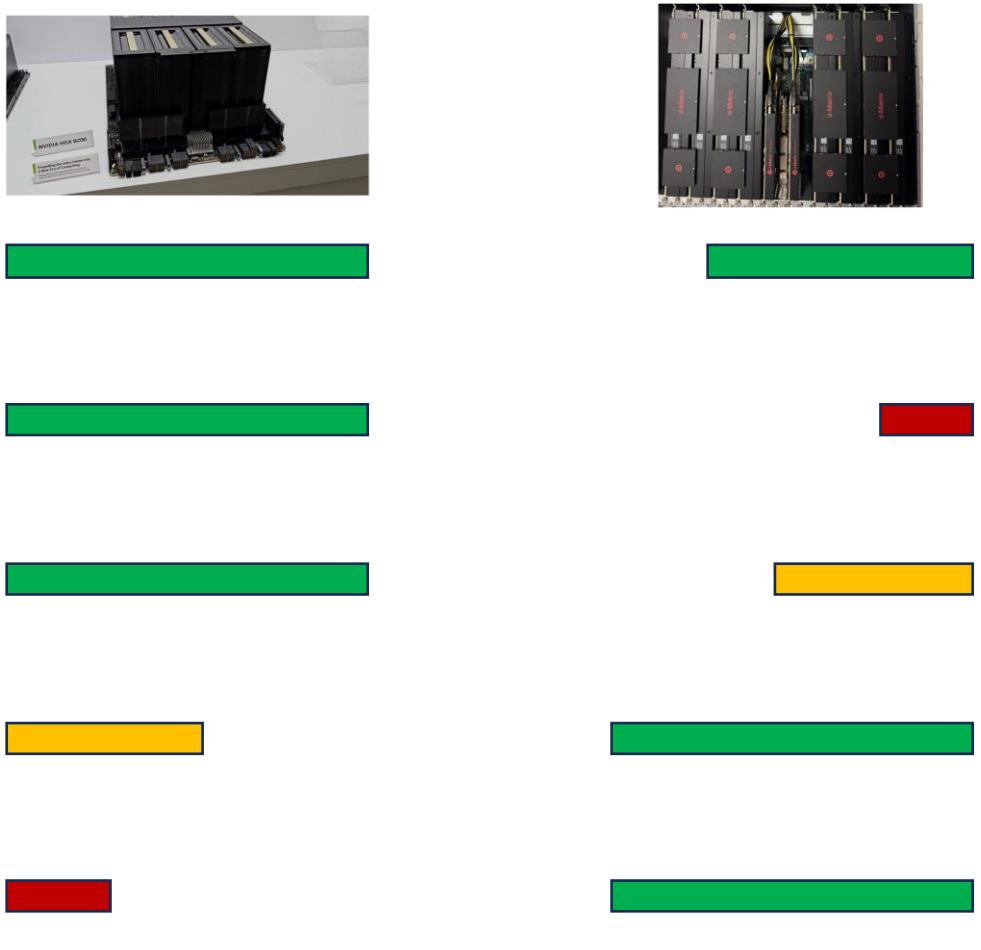
Inference is not one computation — at several levels we see opposite hardware demands.

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A Closer Look at *Corsair* and GPUs

- Corsair shines on
  - Memory bandwidth
  - Connectivity
  - Energy-efficiency
- Falls short on
  - Memory capacity
  - Resultant scaling

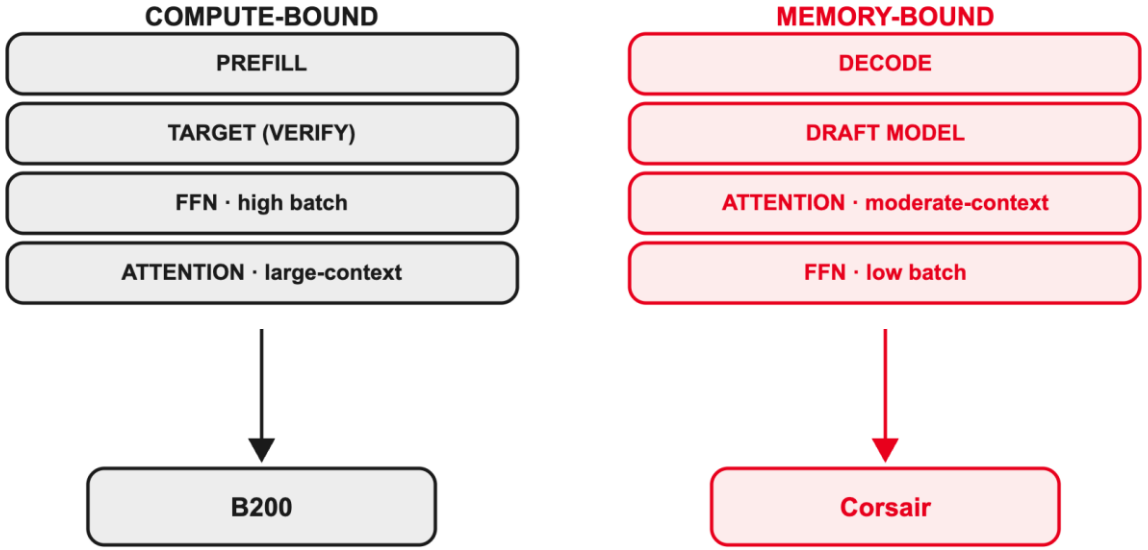
Why Compare?



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Heterogeneous Disaggregated Computing

- Match workload requirements to device capabilities
- Host-free direct DMA between GPU and Corsair over JetStream NICs
- Complementary across a range of deployments

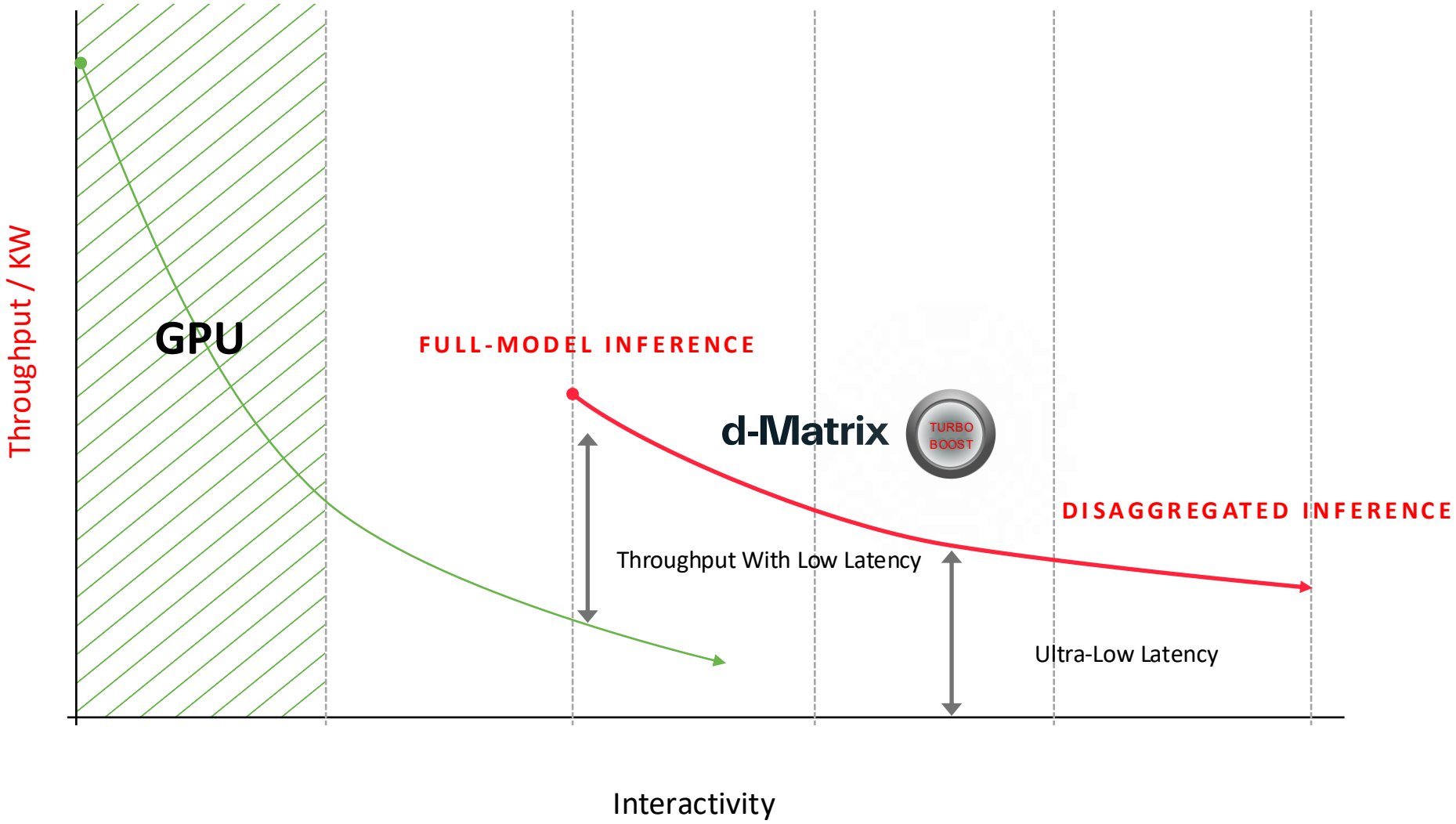




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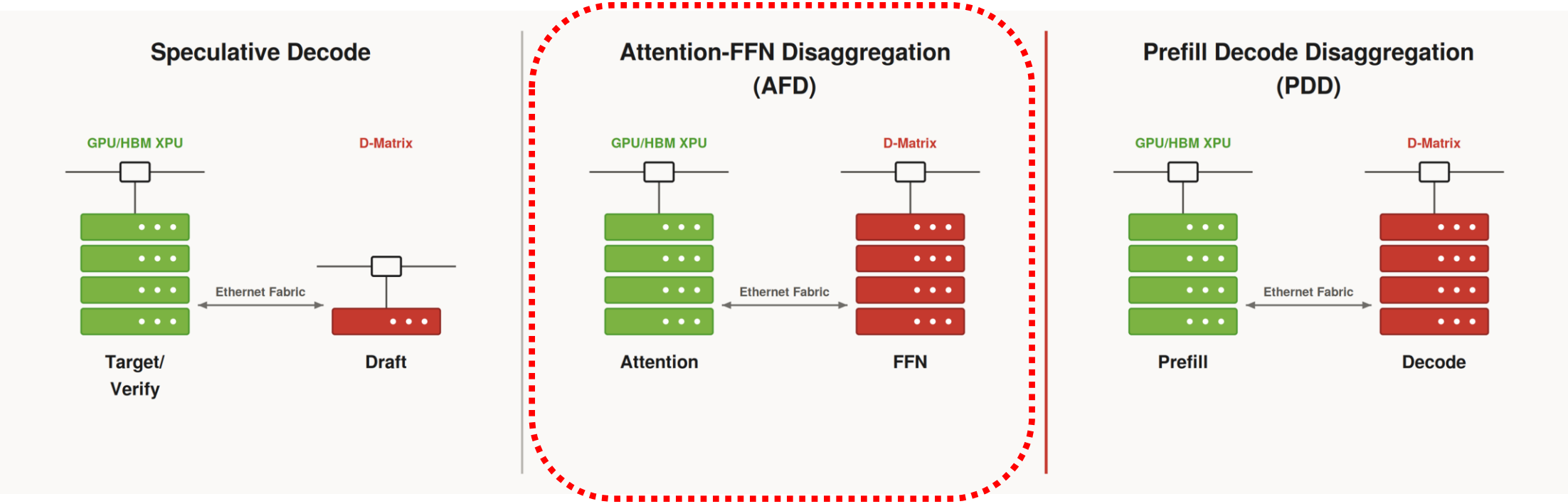
Enabling the Premium (Fast) Token Economy

d-Matrix Gives Boost Needed for Higher Interactivity



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Heterogeneous Compute: Disaggregated Fast LLM



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Heterogeneous AFD

One token, 24 round trips

GPT-OSS-20B · 1× H100 + 8× CORSAIR · BATCH SIZE 1

845 tokens Per Second Per User
64 Concurrent Users = 20,145 tokens/second

Come by Booth C82 for real time demo
@ Wednesday 2 pm PST
@ Thursday 11 am PST

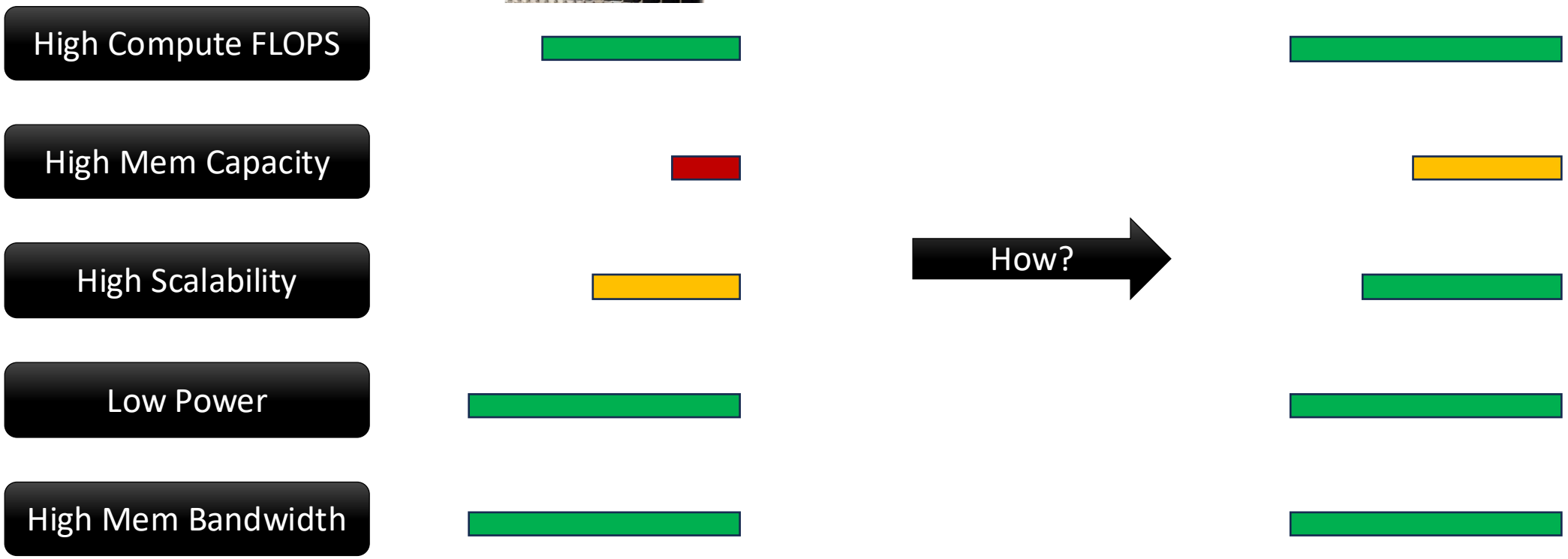
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Quest for the Ideal Inference Accelerator

d-Matrix Corsair

bandwidth-optimal, digital in-memory compute





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Stacked Logic + Memory as the Key Technology

SRAM

On-die memory



Compute + SRAM per tile

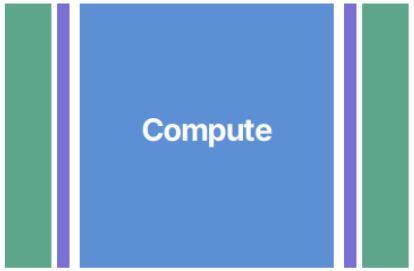
Capacity Low

Bandwidth High

Power Low

HBM

2D package + beachfront



NoC feeds edge HBM stacks

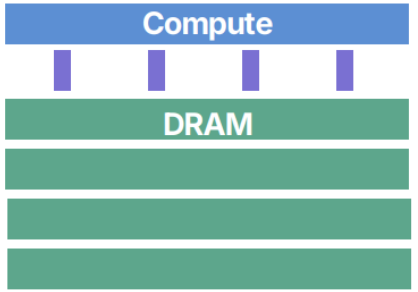
Capacity High

Bandwidth Low

Power High

3D DRAM

Stacked compute-on-memory



Compute on DRAM, vertical IO

Capacity Medium

Bandwidth High

Power Low-Med

The energy ladder

Memories & interconnect	Energy / bit
SRAM (on-die)	**~50 fJ**
On-chip wire	**~35 fJ/mm**
3D vertical IO	**0.3–0.4 pJ**
Interposer trace	**~500 fJ/mm**
2.5D HBM4 (system)	**2.5pJ + 3pJ (on chip)**

3D IO lands **~10× below HBM** — PHY-less, mm-scale vertical path instead of cm-scale interposer + PHY

3D DRAM combines bandwidth and efficiency of SRAM with density of DRAM

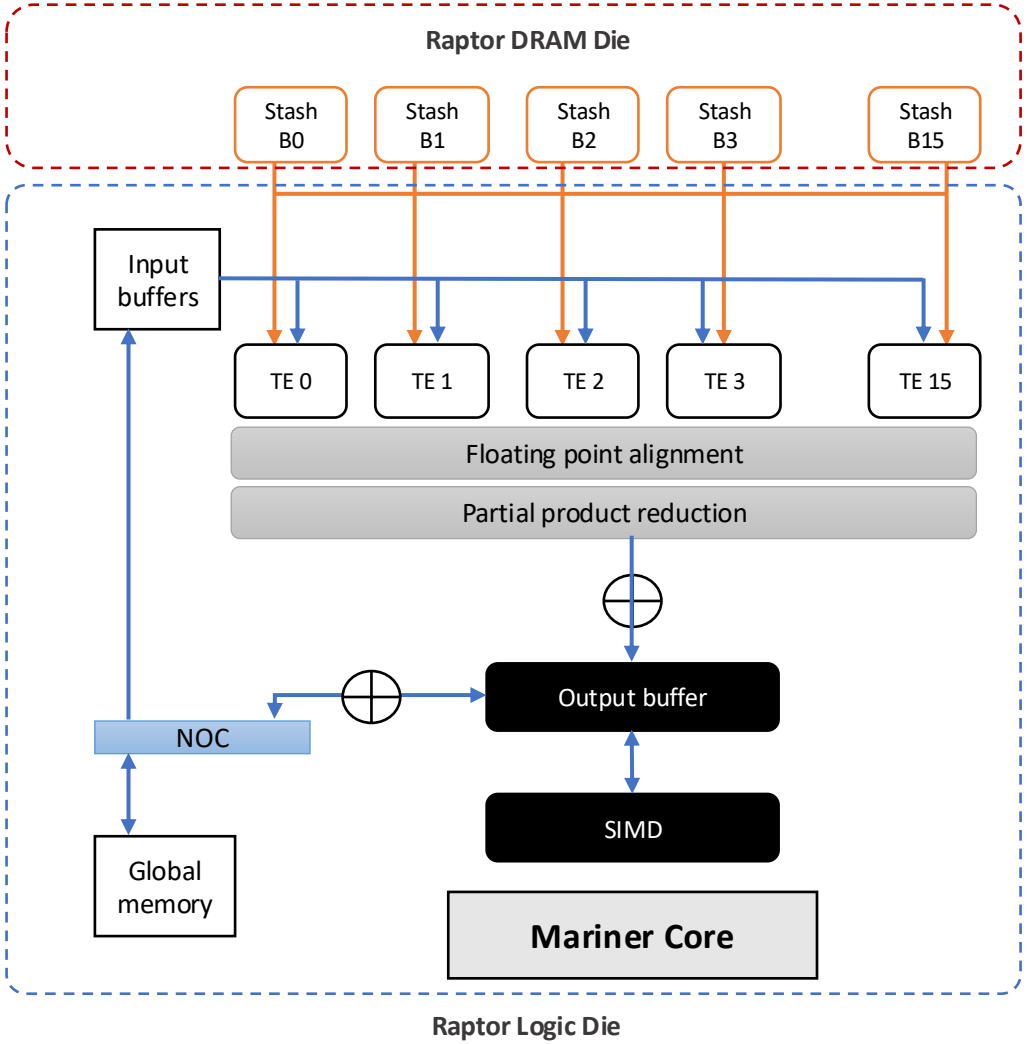
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Raptor Brings DRAM Capacity at SRAM Bandwidths

- Same distributed memory arch as Corsair, just stacked
  - Still aggregates bandwidth
  - Short vertical transfer distances
- Nearly 16 MB backing memory for all 2048 tensor engines

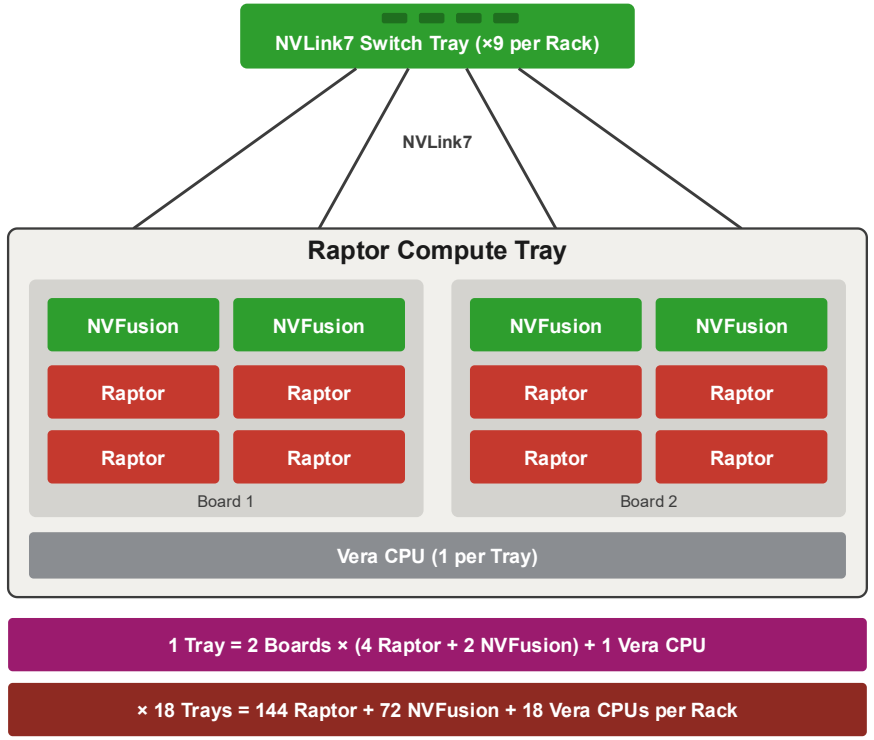
- Raptor increases memory capacity, FLOPS, and scale-up range

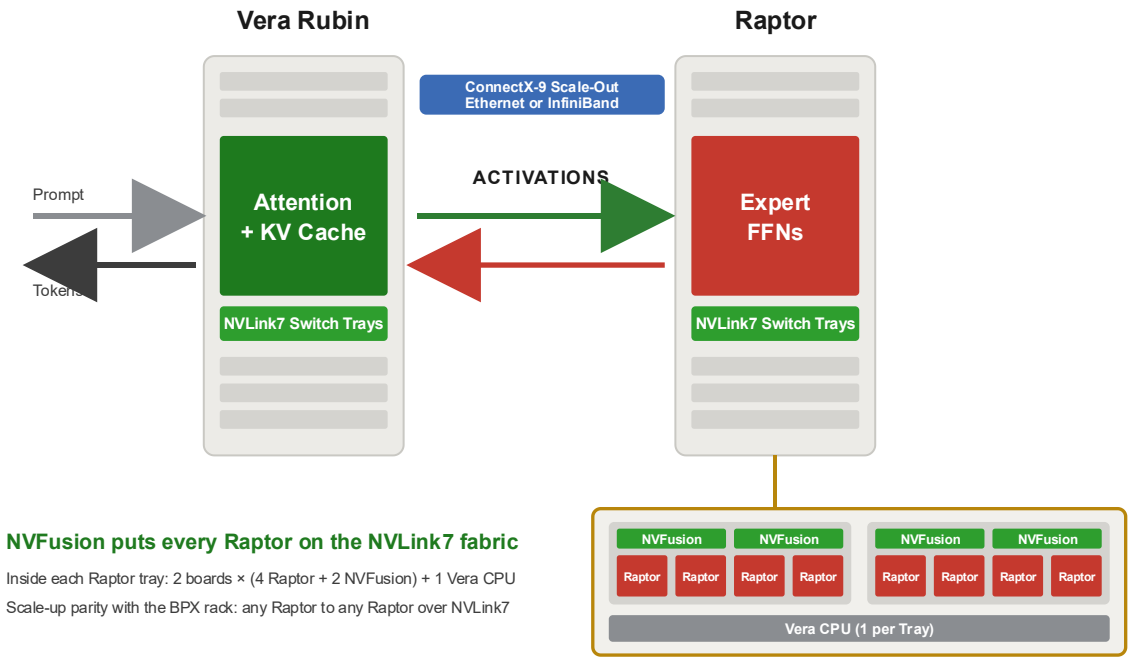




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Scale Up (NVL Fusion) and Scale Out for Raptor



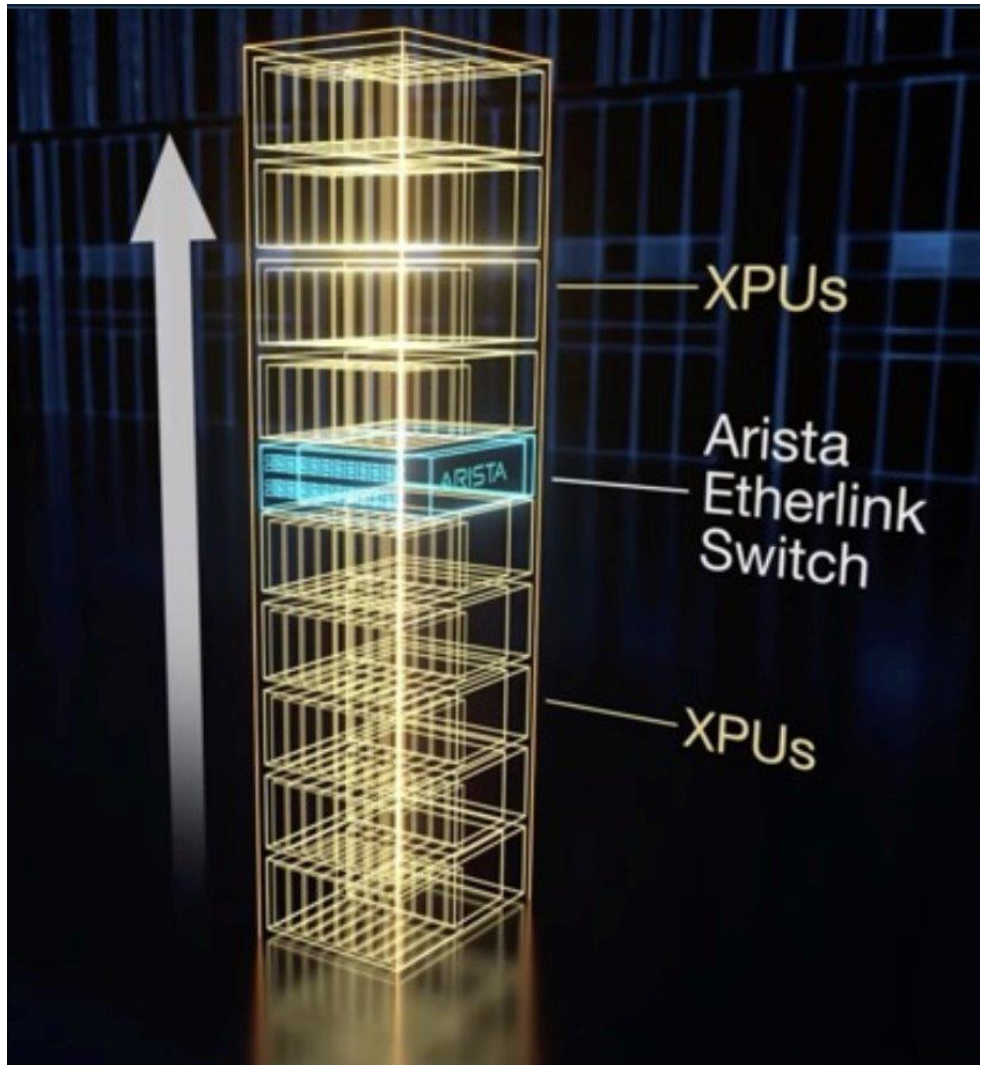


- NVL Fusion for Scale-Up Communication Across 144 Raptors inside a Rack
- Enables frontier model inference on one NVL72 MGX rack

- RDMA communication through IB or Ethernet for Heterogeneous Disaggregation with Vera Rubin

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Scale Up (ESUN)



d-Matrix | ARISTA

Arista AI Networking

ESUN System with 128 Next-Generation Raptor XPUs

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